Pharm D Student's Knowledge, Perception, and Practice of ChatGPT in Clinical Training: A Web-based Cross-sectional Survey in India

Abstract

The emergence of Large Language Models (LLMs) like ChatGPT captured significant attention in healthcare and pharmacy education, yet their real applicability and validity require extensive investigation in practice. Our study aimed to assess Pharm-D students' knowledge, perceptions, and practices (KPP) toward ChatGPT in clinical pharmacy practice training in India. A nationwide, web-based, cross-sectional survey was conducted between June and September 2025 to assess KPP towards ChatGPT use among PharmD students. A self-administered, pre-designed, and validated questionnaire was utilized to collect demographics, educational profiles, and KPP toward ChatGPT use in clinical training. We used social media platforms and Messenger applications and approached professional groups to recruit the study participants by using the snowball sampling technique. A chi-square test was applied to elucidate factors associated with KPP of ChatGPT use in the clinical training. The two-tailed P-value less than 0.05 was considered as a statistically significant value. In our study, the majority ($> 90\%$) of the PharmD students answered all basic knowledge questions about ChatGPT, except the source of ChatGPT's knowledge. The majority (85.8%) perceived

that they indeed benefited from the use of ChatGPT in their training; the agreement levels for benefits varies across different activities. Regarding concerns, ChatGPT use can reduce the interaction with mentors (74.4%), increase the similarity index/AI detection score (64.1%), and increase the risk of inaccurate or misleading information (66.1%), ethical issues (63.3%), and limited applicability (61.0%). The use of ChatGPT in the selection of medicine and calculation of dose was low compared with other clinical pharmacy activities. Variables such as previous experience using AI tools and prior training on AI tool use were significantly associated with greater practice of ChatGPT. Conversely, the perceived benefits of ChatGPT were significantly positively associated with students who had completed internships ($P = 0.019$) and had received prior training ($P = 0.022$) on AI tools. There was a significant positive weak correlation between knowledge and perceived concerns ($r = 0.177$; $P < 0.001$), and moderate positive correlation between perceived benefits and practices of the participants ($r = 0.377$; $P < 0.001$) towards ChatGPT use. Perceived concerns ($r = -0.126$; $P < 0.012$) were significantly negatively weakly correlated with ChatGPT use practices in clinical training. Majority of the PharmD students have good knowledge about ChatGPT, and positive perception towards ChatGPT benefits in clinical training. However, the practice was limited to certain activities due to concerns about accuracy, ethics, and reduced mentor interaction upon use of ChatGPT. Prior experience and training were significantly associated with high practice and positive attitudes. The study recommends to integrate AI tool use and application in pharmacy curriculum and train the faculty to promote competent and responsible use of ChatGPT in clinical pharmacy practice. Students must validate the information from the standard resources or interacting with mentors to deal with misleading information generated by AI tools.